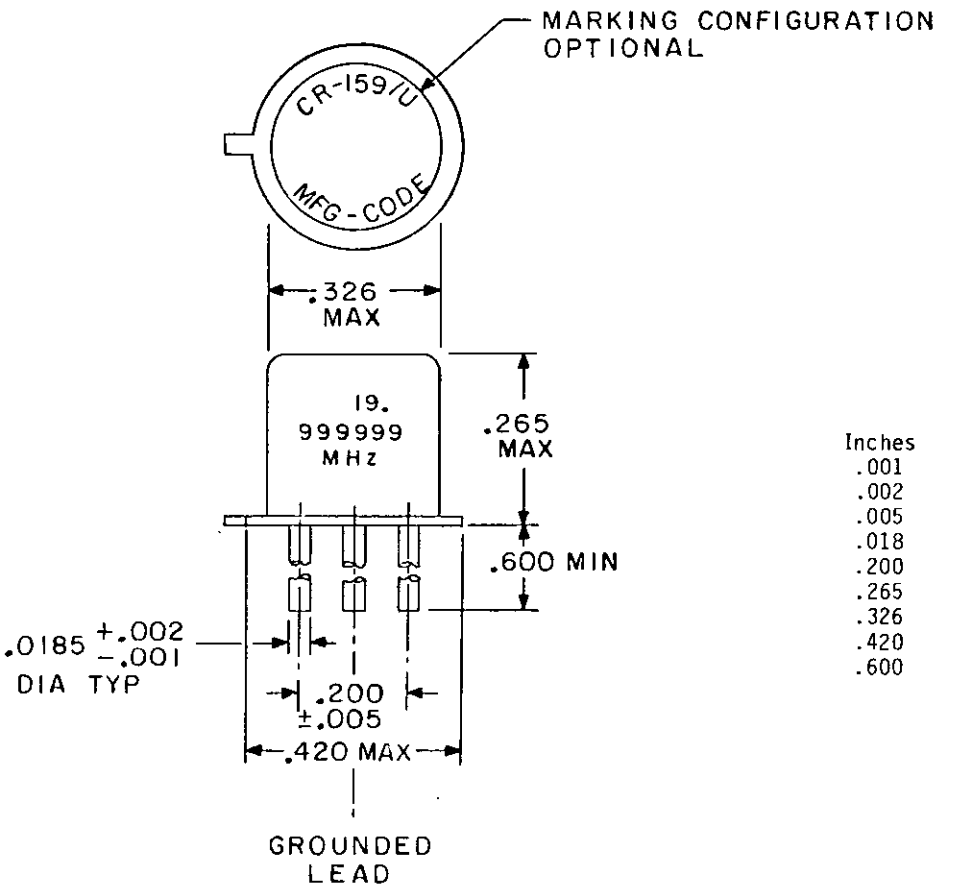
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-159/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 8 to 20 MHz; fundamental; noncontrolled; series.



Inches	mm
.001	0.02
.002	0.05
.005	0.12
.018	0.46
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The crystal unit shall not be trimmed after sealing.

FIGURE 1. Crystal unit - CR-159/U.

Ⓓ denotes changes

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TABLE I. Requirements.

Requirements	Specified value or condition
Holder	HC-35/U (MIL-H-10056/21)
Dimensions, marking, and configuration	See figure 1
Frequency range	8 to 20 MHz, inclusive
Frequency tolerance	±50 ppm
Equivalent resistance	See table III
Mode of oscillation	Fundamental
Operating temperature range	-55°C to +105°C, inclusive
Test set, calibration values and rated drive level	See table II
Capacitance, shunt	7 pF, maximum
Shock (specified pulse):	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±10 ppm
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Low temperature storage:	
Resistance	See table III
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	8+ to 10	16	11	---	2.0 ±1.0
	10+ to 15	13	12	---	2.0 ±1.0
TS-683/TSM	15+ to 20	12	---	0.15	2.0 ±1.0

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TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
8+ to 10	40
10+ to 12	30
12+ to 15	25
15+ to 20	20

(D) Qualification inspection reinstated as of the date of this document.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0645)

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